Student Name :
Student ID:

<u>Choose the correct answer:</u>

Q1

A university computer lab has hundreds of students using the same server simultaneously. The operating system must ensure that each student receives fair access to the CPU and memory.

The operating system is acting primarily as a:

A. Device Driver

B. Application Manager

C. **Resource Allocator**

D. Compiler

✅ **Answer: C**

Q2

A user clicks **Print** in Microsoft Word. Before the printer receives the request, the operating system is involved because it:

A. Converts the document into binary.

B. **Acts as an intermediary between applications and hardware.**

C. Increases the printer's memory.

D. Restarts the CPU.

✅ **Answer: B**

Q3

A cloud server receives thousands of requests from different users. The operating system must decide which process gets CPU time first.

This demonstrates the OS goal of:

A. **Efficient resource utilization**

B. Disk formatting

C. Virus detection

D. Data compression

✅ **Answer: A**

Q4

A software engineer installs Google Chrome on a Windows computer.

Google Chrome is considered:

A. Kernel

B. Firmware

C. Hardware

D. **Application Software**

✅ **Answer: D**

Q5

A payroll application needs memory, CPU time, and access to storage.

Who manages these resources?

A. BIOS

B. Device Controller

C. **Operating System**

D. Cache Memory

☑ **Answer: C**

Q6

A smartwatch is designed to consume very little power while providing basic functionality.

This is an example of:

A. Supercomputer

B. **Handheld Computer**

C. Mainframe

D. Workstation

☑ **Answer: B**

Q7

Immediately after pressing the power button, which program begins execution first?

A. **Bootstrap Program**

B. Device Driver

C. User Application

D. Operating System Shell

☑ **Answer: A**

Q8

The bootstrap program is normally stored inside:

A. RAM

B. Cache

C. SSD

D. **ROM/Firmware**

✅ **Answer: D**

Q9

Which task is performed by the bootstrap program during startup?

A. Opens user applications.

B. **Loads the operating system kernel after initializing hardware.**

C. Deletes temporary files.

D. Starts the web browser.

✅ **Answer: B**

Q10

A keyboard finishes sending data to the computer.

How does it notify the CPU?

A. Polling

B. DMA

C. **Interrupt**

D. Scheduling

✅ **Answer: C**

Q11

A printer finishes printing a report and informs the processor that the job is complete.

This notification is called:

A. **Interrupt**

B. Trap

C. Thread

D. Context Switch

✅ **Answer: A**

Q12

While servicing one interrupt, the operating system temporarily blocks new interrupts.

Why?

A. To reduce RAM usage.

B. To increase cache speed.

C. To save battery power.

D. **To prevent losing another interrupt.**

✅ **Answer: D**

Q13

A programmer accidentally executes:

```
int x = 10 / 0;
```

This causes:

A. Hardware Interrupt

B. Timer Interrupt

C. **Software Interrupt (Trap)**

D. Device Interrupt

✅ **Answer: C**

Q14

Immediately after an interrupt occurs, what does the operating system save first?

A. Hard disk sectors

B. **CPU registers and the program counter**

C. Cache contents

D. BIOS configuration

✅ **Answer: B**

Q15

A server repeatedly checks whether a network card has received new data instead of waiting for an interrupt.

This technique is called:

A. **Polling**

B. Scheduling

C. Paging

D. Swapping

☑ **Answer: A**

Q16

A company upgrades its database servers by replacing HDDs with SSDs to reduce the time required to retrieve frequently accessed records.

What is the primary advantage of SSDs?

A. Lower storage capacity

B. **Faster access time**

C. Volatile storage

D. Larger cache memory

☑ **Answer: B**

Q17

A processor checks a small, high-speed memory before accessing the main memory. If the required data is found there, execution continues immediately.

Which storage component is being used?

A. Main Memory (RAM)

B. Hard Disk

C. Registers

D. **Cache Memory**

☑ **Answer: D**

Q18

A web browser copies frequently accessed web pages into a faster storage area so that they can be loaded more quickly during future visits.

Which operating-system principle is being applied?

A. Interrupt Handling

B. Virtualization

C. **Caching**

D. Swapping

✅ **Answer: C**

Q19

During a sudden power outage, all currently running programs disappear, but files stored on the SSD remain available after power is restored.

Which storage component is volatile?

A. **Main Memory (RAM)**

B. SSD

C. Hard Disk

D. Magnetic Tape

✅ **Answer: A**

Q20

A university archive stores students' records for many years, even when computers are powered off.

Which storage type should be used?

A. Cache

B. **Secondary Storage**

C. CPU Registers

D. RAM

☑ **Answer: B**

Q21

A company installs four identical processors, and each processor can execute any operating-system task independently.

This architecture is called:

A. Asymmetric Multiprocessing

B. Single Processing

C. Time Sharing

D. **Symmetric Multiprocessing (SMP)**

☑ **Answer: D**

Q22

In a network router, one processor is dedicated to security monitoring while another handles packet forwarding.

This architecture represents:

A. **Asymmetric Multiprocessing (AMP)**

B. Symmetric Multiprocessing

C. Batch Processing

D. Multiprogramming

☑ **Answer: A**

Q23

An employee edits a report while a large file is downloading in the background. Whenever the download waits for network input, the operating system allows the editor to continue running.

Which concept is illustrated?

A. Bootstrapping

B. **Multiprogramming**

C. Polling

D. Interrupt Vector

☑ **Answer: B**

Q24

A software developer listens to music, participates in a video meeting, and edits source code simultaneously on the same computer.

This interactive capability is provided by:

A. Batch Processing

B. Boot Loader

C. **Time Sharing (Multitasking)**

D. ROM

☑ **Answer: C**

Q25

A running application must wait for data from a slow storage device. Instead of leaving the CPU idle, the operating system immediately executes another ready program.

The primary benefit is:

A. Larger cache size

B. Increased disk capacity

C. Faster boot time

D. **Higher CPU utilization**

✅ **Answer: D**

Q26

A spreadsheet application requests the operating system to create a new file on the disk.

This request is handled through a:

A. Hardware Interrupt

B. DMA Transfer

C. **Software Interrupt (System Call/Trap)**

D. Cache Miss

✅ **Answer: C**

Q27

A malicious application attempts to modify memory that belongs to the operating system.

Which mechanism prevents this action?

A. **Dual-Mode Operation (User/Kernel Mode)**

B. Cache Replacement

C. Multiprogramming

D. Device Scheduling

✅ **Answer: A**

Q28

A device driver needs to execute privileged CPU instructions while communicating directly with hardware.

Which execution mode is required?

A. User Mode

B. **Kernel Mode**

C. Safe Mode

D. Virtual Mode

☑ **Answer: B**

Q29

After the operating system completes a requested service (system call), where does execution continue?

A. BIOS

B. Kernel Scheduler

C. Boot Loader

D. **User Mode**

☑ **Answer: D**

Q30

A student accidentally writes an infinite loop that continuously consumes CPU time. The operating system regains control after a predefined time interval.

Which hardware feature makes this possible?

A. Cache Memory

B. Device Driver

C. **Timer**

D. Virtual Memory

☑ **Answer: C**